



Cambridge IGCSE™

COMBINED SCIENCE

Paper 1 Multiple Choice (Core)

0653/13

May/June 2026

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

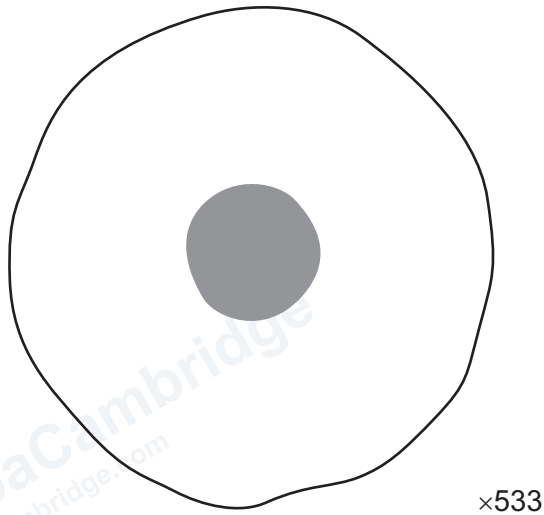
- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

- 1 The diagram shows a magnified human ovum.

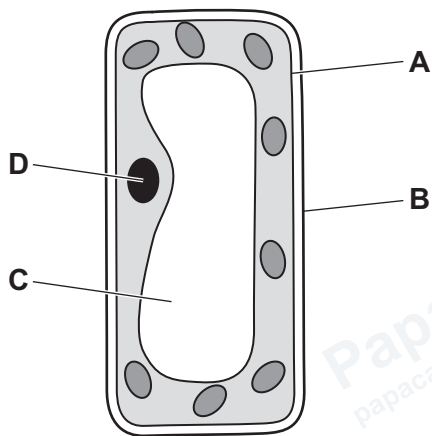


The ovum in the diagram measures 64 μm in diameter.

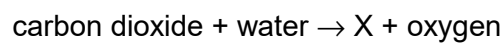
What is the actual diameter of the ovum?

- A** 0.012 mm **B** 0.12 mm **C** 0.83 mm **D** 8.30 mm

- 2 Which part of a cell contains the genetic information?



- 3 The equation for photosynthesis is shown.



What does X represent?

- A** carbon
B chlorophyll
C glucose
D light

4 Which row describes the effects of physical digestion on food?

	effect on size of food pieces	chemical change occurs
A	large to small	no
B	large to small	yes
C	small to large	no
D	small to large	yes

5 Which pathway is taken by water through a plant?

- A mesophyll cells → xylem → root cortex cells → root hair cells
- B root cortex cells → root hair cells → mesophyll cells → xylem
- C root hair cells → root cortex cells → xylem → mesophyll cells
- D root hair cells → xylem → root cortex cells → mesophyll cells

6 What is a type of blood vessel that carries blood from the heart to the body?

- A atria
- B artery
- C capillary
- D vein

7 What are body defences against disease?

- 1 hairs in nose
- 2 hairs on skin
- 3 red blood cells
- 4 white blood cells

- A 1 and 3
- B 1 and 4
- C 2 and 3
- D 2 and 4

- 8 Which row describes the changes to the depth and rate of breathing when a person starts to exercise?

	depth of breathing	rate of breathing
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

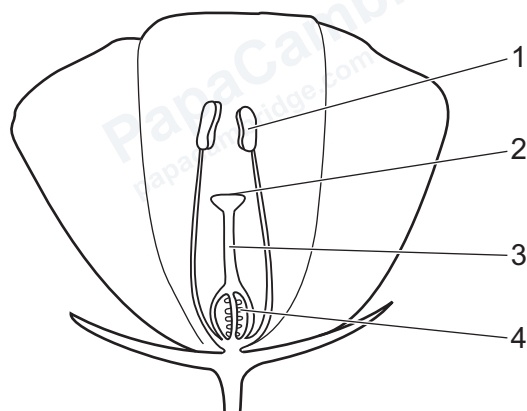
- 9 Which process uses energy released from glucose?

- A** diffusion of oxygen
- B** osmosis of water
- C** synthesis of enzymes
- D** transpiration of water vapour

- 10 Which term is used to describe antibiotics?

- A** drugs
- B** enzymes
- C** hormones
- D** nutrients

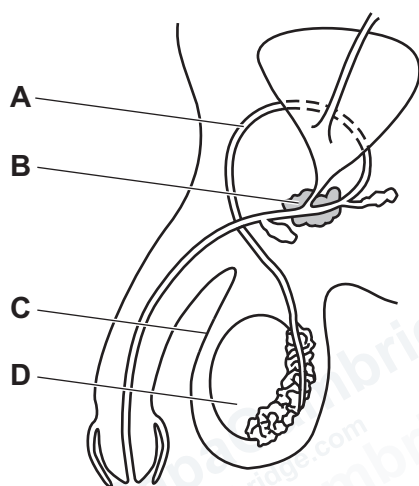
11 The diagram shows a section through an insect-pollinated flower.



Which row identifies the sites of pollen production, pollination and fertilisation?

	pollen production	pollination	fertilisation
A	1	2	4
B	1	3	4
C	2	3	1
D	4	2	3

12 The diagram shows the male reproductive system.



Which row identifies the labelled structure and states its function?

	name of structure	function of structure
A	sperm duct	sperm pass directly into the uterus from this tube
B	testis	produces the sperm
C	scrotum	holds the testis
D	prostate gland	stores the sperm

13 What is the principal source of energy in biological systems?

- A food
- B plants
- C respiration
- D the Sun

14 Salt, sand and water are stirred together in a beaker.

The salt dissolves in the water.

What does the beaker contain?

- A a mixture of a solution and a solid
- B a mixture of three elements
- C only one compound and one solid
- D only one compound containing three elements

15 Which statement about elements in the same period of the Periodic Table is correct?

- A Elements in the same period have similar chemical properties.
- B The elements become more metallic across the period.
- C The number of electron shells in an atom increases across the period.
- D The number of outer-shell electrons in an atom increases across the period.

16 Which statement about particles is correct?

- A H_2 has the same number of protons as neutrons.
- B H^+ has more electrons than protons.
- C NH_3 has the same number of protons as electrons.
- D OH^- contains more protons than electrons.

17 Which substances undergo electrolysis?

- 1 concentrated aqueous sodium chloride
- 2 distilled water
- 3 molten lead(II) bromide

- A 1 and 2 B 1 and 3 C 2 and 3 D 3 only

18 Which statement describes redox reactions?

- A Redox reactions involve only oxidation.
- B Redox reactions involve only reduction.
- C Redox reactions involve simultaneous oxidation and reduction.
- D Redox reactions involve oxidation followed by reduction.

19 A salt is added to water.

The temperature of the water is measured before the salt is added and after the salt has dissolved completely.

initial temperature /°C	final temperature /°C
22	15

Which word describes this process?

- A condensation
- B crystallisation
- C endothermic
- D exothermic

20 Separate samples of ammonia, carbon dioxide, hydrogen and oxygen are tested with universal indicator.

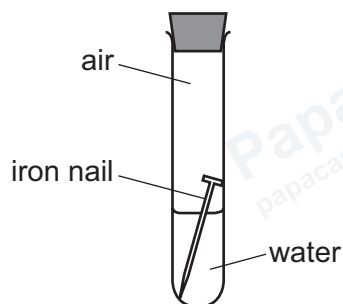
How many of these gases turn the universal indicator orange or red?

- A 1 B 2 C 3 D 4

21 Which row shows the properties of a transition element?

	melting point	electrical conductivity	colour of chloride	catalytic properties
A	high	high	white	no
B	high	low	white	no
C	high	high	green	yes
D	low	low	blue	yes

- 22 An iron nail is left for a few days in a sealed test-tube containing air and water.



Which change to the amount of gases in the air in the test-tube occurs as the iron nail rusts?

- A The amount of carbon dioxide decreases.
 - B The amount of carbon dioxide increases.
 - C The amount of nitrogen increases.
 - D The amount of oxygen decreases.
- 23 Which metal is extracted from its ore using carbon?

- A aluminium
- B copper
- C magnesium
- D potassium

- 24 Four gases are listed.

- 1 carbon dioxide
- 2 nitrogen
- 3 oxygen
- 4 methane

Which gases increase global warming?

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

- 25 Which statements about fossil fuels are correct?

- 1 Natural gas and petroleum are the only two fossil fuels.
- 2 Petroleum contains a mixture of hydrocarbons.
- 3 Petroleum is separated into useful substances by chromatography.
- 4 The main constituent of natural gas is an alkane.

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

26 Which statement about mixtures is correct?

- A A filtrate is a solution that passes through a filter.
- B A residue is a liquid that is removed when the mixture is distilled.
- C A solute is a mixture of a soluble solid and a solvent.
- D A solvent is a solid that can be dissolved in water.

27 Aqueous X is acidified with dilute nitric acid.

Aqueous barium nitrate is then added.

A white precipitate forms.

Which anion does X contain?

- A carbonate
- B chloride
- C nitrate
- D sulfate

28 What is the meaning of the weight of an object?

- A the density of the material from which it is made
- B the force exerted on it by gravity
- C the mass of the matter it contains
- D the pressure it exerts on the ground

29 A measuring cylinder contains 60 cm^3 of water.

A solid object of mass 120 g is lowered into the water until it is completely submerged.

The new reading on the measuring cylinder is 80 cm^3 .

What is the density of the object?

- A 0.50 g/cm^3 B 1.5 g/cm^3 C 2.0 g/cm^3 D 6.0 g/cm^3

30 A crane lifts three loads, X, Y and Z.

- Load X weighs 400 N and is lifted through 15 m.
- Load Y weighs 400 N and is lifted through 12 m.
- Load Z weighs 600 N and is lifted through 12 m.

The work done on each load is calculated.

What is the order of the loads starting with the load with the least work done to the load with the most work done?

A $X \rightarrow Y \rightarrow Z$ **B** $Y \rightarrow X \rightarrow Z$ **C** $Z \rightarrow X \rightarrow Y$ **D** $Z \rightarrow Y \rightarrow X$

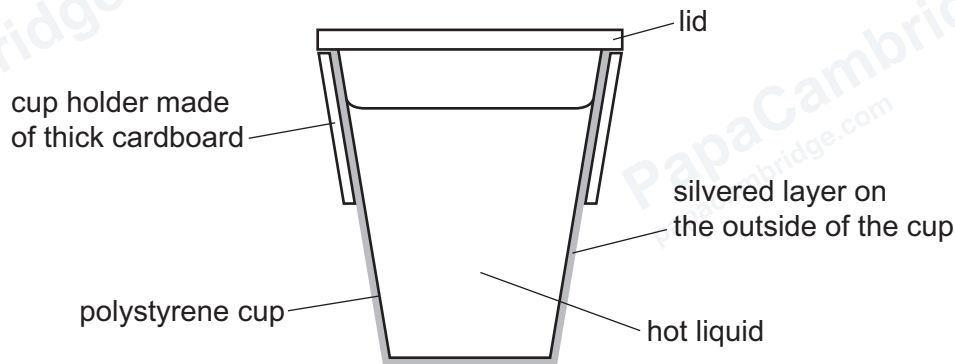
31 Which row gives two properties of a liquid?

	property 1	property 2
A	easy to compress	flows
B	easy to compress	does not flow
C	difficult to compress	flows
D	difficult to compress	does not flow

32 What happens to a metal rod when its temperature increases?

- A** It becomes longer.
- B** It becomes shorter.
- C** It becomes heavier.
- D** It becomes lighter.

- 33 A student designs a cup with features that reduce energy transfer from a hot liquid in the cup to the surroundings.



Which feature only reduces energy transfer by radiation?

- A a silvered layer on the outside of the cup
 - B a lid
 - C a cup holder made of thick cardboard
 - D a cup made of polystyrene
- 34 Red light enters a block of glass and the direction of travel of the light changes.

What is this change of direction called?

- A dispersion
 - B radiation
 - C reflection
 - D refraction
- 35 Radio waves, visible light and X-rays all travel in a vacuum.

Which statement about the waves is correct?

- A Radio waves travel at the greatest speed.
- B Visible light travels at the greatest speed.
- C X-rays travel at the greatest speed.
- D They all travel at the same speed.

- 36 A sound becomes louder and lower pitched.

Which row states how the amplitude and the frequency of the sound change?

	amplitude	frequency
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 37 A student measures the current in a circuit, the voltage of the power supply and the time for which the circuit is switched on.

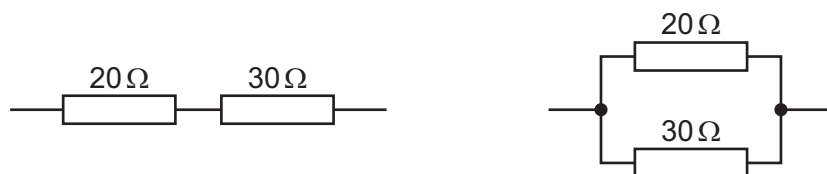
Which quantities are needed to calculate the energy transferred?

- A** the current, the voltage and the time
- B** the current and the voltage only
- C** the current and the time only
- D** the voltage and the time only

- 38 What is meant by a kilowatt-hour (kWh)?

- A** the energy transferred in 1 hour by an appliance with a power of 1 kW
- B** the power of an appliance that transfers 1 kW of energy in 1 hour
- C** the time taken for an appliance with a power of 1 kW to transfer 1 J of energy
- D** the time taken for an appliance to transfer 1 kW of energy

- 39 A $20\ \Omega$ and a $30\ \Omega$ resistor are connected first in series and then in parallel, as shown.



Which row describes the combined resistance of the two resistors in series and in parallel?

	combined resistance in series	combined resistance in parallel
A	between $20\ \Omega$ and $50\ \Omega$	$50\ \Omega$
B	less than $20\ \Omega$	$50\ \Omega$
C	$50\ \Omega$	less than $20\ \Omega$
D	$50\ \Omega$	between $20\ \Omega$ and $50\ \Omega$

- 40 Which row contains two correct statements about the Solar System?

	statement 1	statement 2
A	Pluto is a dwarf planet.	An asteroid orbits a planet.
B	Pluto is a dwarf planet.	A moon orbits a planet.
C	The Sun orbits the Earth.	An asteroid orbits a planet.
D	The Sun orbits the Earth.	A moon orbits a planet.

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The Periodic Table of Elements

Group																	
I	II	Group														VII	VIII
		1 H hydrogen 1															2
		3 Li lithium 7															He
		11 Na sodium 23															Ne
		19 K potassium 39															Ar
		37 Rb rubidium 85															Kr
		55 Cs caesium 133															Xe
		87 Fr francium —															Rn
		21 Sc scandium 45															
		39 Y yttrium 89															
		57–71 lanthanoids															
		89–103 actinoids															
		23 V vanadium 51															
		22 Ti titanium 48															
		21 Sc scandium 45															
		20 Ca calcium 40															
		19 K potassium 39															
		18 Ar argon 40															
		17 Cl chlorine 35.5															
		16 S sulfur 32															
		15 P phosphorus 31															
		14 Si silicon 28															
		13 Al aluminium 27															
		12 Mg magnesium 24															
		11 Na sodium 23															
		10 Ne neon 20															
		9 F fluorine 19															
		8 O oxygen 16															
		7 N nitrogen 14															
		6 C carbon 12															
		5 B boron 11															
		4 Be beryllium 9															
		3 Li lithium 7															
		2 He helium 4															

lanthanoids	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175
	89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).